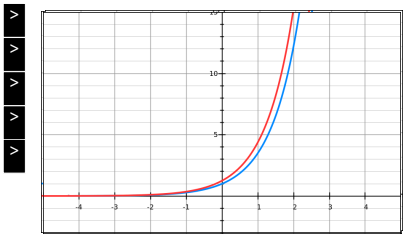
1.23 e and Natural Logarithms

Chapter 14 · Original slide 8 · Exponentials and logarithms



Click

You won't yet be able to differentiate general exponential functions $y = a^x$ until Year 2.

But I've calculated the gradient functions for you – click the black arrow to reveal the graph and gradient function.

Function

Gradient

$y = 1^x$

$\frac{dy}{dx} = 0$

>

$y = 1.5^x$

$y = 2^x$

$\frac{dy}{dx} = 0.41 \times 1.5^x$

$\frac{dy}{dx} = 0.69 \times 2^x$

$y = 2.5^x$

$\frac{dy}{dx} = 0.92 \times 2.5^x$

$y = 3^x$

$\frac{dy}{dx} = 1.10 \times 3^x$

$y = 3.5^x$

$\frac{dy}{dx} = 1.25 \times 3.5^x$

Compare each exponential function against its respective gradient function. What do you notice?

Chapter 14 · Original slide 9 · y={e}^{x}

Function

Gradient

$y = 1^x$

$y = 2.5^x$ and $y = 3^x$ seem to be similar to their respective gradient functions. So is there a base between 2.5 and 3 where the function is equal to its gradient function?

$e = 2.71828\ldots$ is known as Euler's Number.

It is one of the five most fundamental constants in mathematics (0, 1, i, e, π).

It has the property that:

$\frac{dy}{dx} = 0$

>

$y = 1.5^x$

$y = 2^x$

$\frac{dy}{dx} = 0.41 \times 1.5^x$

$\frac{dy}{dx} = 0.69 \times 2^x$

>

>

$y = 2.5^x$

$\frac{dy}{dx} = 0.92 \times 2.5^x$

>

$y = 3^x$

$\frac{dy}{dx} = e^x$

$\frac{dy}{dx} = 1.10 \times 3^x$

1.23 e and Natural Logarithms

Chapter 14 · Original slide 10 · Differentiating y=a{e}^{kx}

! If $y = e^{kx}$, where k is a constant, then $\frac{dy}{dx} = ke^{kx}$

Note: This is not a standalone rule but an application of something called the '*chain rule*', which you will encounter in Year 2.

$y = e^{-1x} \quad \therefore \quad \frac{dy}{dx} =$

Note: In general, when you scale the function, you scale the derivative/integral.

Different e^{5x} with respect to x.

Different e^{-x} with respect to x.

Different $4e^{3x}$ with respect to x.

Chapter 14 · Original slide 41 · Natural Logarithms

We have previously seen that $y = \log_a x$ is the inverse function of $y = a^x$.

We also saw that e^x is "the" exponential function.

The inverse of e^x is $\log_e x$, but because of its special importance, it has its own function name!

! The inverse of $y = e^x$ is $y = \ln x$

$\ln e^x =$

$e^{\ln x} =$

Since "e to the power of" and "ln of" are inverse functions, they cancel each other out!

Solve $e^x = 5$

"ln both sides".

On the LHS it cancels out the "e to the power of"

Solve $2 \ln x + 1 = 5$

Do "e to the power of" each side. On the LHS it cancels out the ln.

1.23 e and Natural Logarithms

Chapter 14 · Original slide 42 · Quadratics in {e}^{x}

In previous chapters we've already dealt with quadratics in disguise, e.g. "quadratic in sin". We therefore just apply our usual approach: either make a suitable substitution so the equation is then quadratic, or (strongly recommended!) go straight for the factorisation.

Solve $e^{2x} + 2e^{x} - 15 = 0$

Solve $e^{x} - 2e^{-x} = 1$

First write negative powers as fractions:

Note that $e^{2x} = (e^{x})^{2}$ therefore

Chapter 14 · Original slide 43 · Test Your Understanding

Solve $\ln(3x + 1) = 2$

Solve $e^{2x} + 5e^{x} = 6$

Solve $2^{x}e^{x+1} = 3$ giving your answer as an exact value.

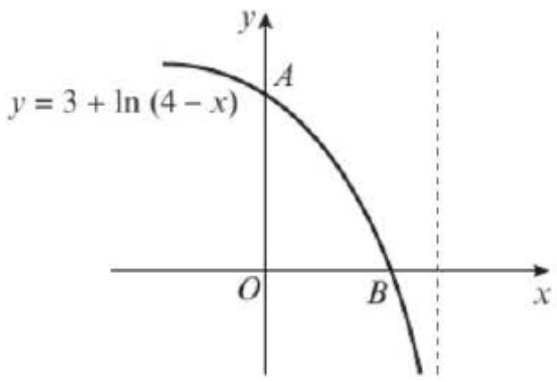
1.23 e and Natural Logarithms

Chapter 14 · Original slide 44 · Exercise 14G – Y1 Page 327

4 Find the exact solutions to the equation $e^x + 12e^{-x} = 7$.

6 Solve $3^x e^{4x-1} = 5$, giving your answer in the form $\dfrac{a + \ln b}{c + \ln d}$

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8 The graph of $y = 3 + \ln(4 - x)$ is shown to the right.

a State the exact coordinates of point A. (1

b Calculate the exact coordinates of point B. (3 m

Challenge: The graph of the function passes through (0,5) and (6,10). Given that the line $y = 2$ is an asymptote to the graph, show that $B = \frac{1}{6}\ln\left(\frac{8}{3}\right)$